National University of Computer and Emerging Sciences

Technical & Business Writing (2007)

Date: February 28 2024

Course Instructor(s)

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Sessional-I Exam

Total Time: 1 Hour
Total Marks: 30
Total Questions: **02**

Semester: SP-2024
Campus: Karachi
Dept: Computer Science

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Student Name	Roll No	Section	Student Signature

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CLO #4: Displays the internationally accepted standards of technical communication.

Q1: Do as directed. **[20 marks]**

a. The following paragraph has vague information and words. Re-write it by providing relevant details and information. [5]

In this project, we will create a secure online auction system using the fraud detection method with binary classification. If a user wants to buy a product through an online auction, they must provide some basic information about themselves. The system will then screen the users, authenticate, and authorize them. Only authorized users can bid in the auction. Buying will become hassle free an easy for everyone. Users will also easily get all useful information about whatever they desire on the homepage.

b. Rewrite the following sentences improving economy and conciseness using the strategies taught in the course. [2]

1. It is a commonly accepted fact that the identification and fixing of bugs early in the development process can be effectively done through the practice of unit testing, which is a software testing technique involving the testing of individual units or components of a software application in isolation.

2. Code refactoring, a practice that involves restructuring existing computer code without changing its external behavior, is something that software developers often engage in to improve code readability and maintainability in the long run.

c. The following sentences contain unnecessary nominalizations. Rewrite them by liberating the disguised verbs. [3]

1. The establishment of a connection between the client and the server is a prerequisite for the initiation of data transfer.
2. The implementation of updates is necessary for the enhancement of system security.
3. The execution of a test suite is crucial for the validation of software functionality.

d. The table below contains statements taken from a technical report on "Unmanned Aerial Vehicles". Read them and identify the target readers of each set of sentences. You can choose from the options below. [5]

Researchers/Governmental Bodies/General Public/Industry Executive

Set No.	Sentence(s)	Target Reader
1	Applications of unmanned aerial vehicles include aerial photography, precision agriculture, forest fire monitoring, river monitoring, environmental monitoring, policing and surveillance, infrastructure inspections, smuggling, and product deliveries.	
2	The first aircraft autopilot was developed by Sperry Corporation in 1912 [6] and demonstrated in a hands-free flight two years later.	
3	A UAV autopilot system is a close-loop control system, which comprises of two parts: the state observer and the controller. The most common state observer is the micro inertial guidance system including gyro, acceleration, and magnetic sensors.	
4	Adaptive neural network controller does not require an accurate mathematical model and is suitable for multi-variable flight control.	
5	Many organizations see drones as an economical and more efficient mode for product delivery as drones avoid traffic on the road, thus providing faster delivery.	

e. Study the following pairs of sentences. For each pair, one is more objective, while the other is less suitable for academic writing. Identify which ones are more objective. [5]

Pair. No	A	B
1	First, the temperature of the flask was measured.	First, I measured the temperature of the flask.
2	I have highlighted three problem areas.	This report has highlighted three problem areas.
3	We can see how the rate increased in the graph.	The graph shows how the rate increased.
4	There are two problems with this method	This is a really bad method.
5	Kendrick (2020) advises against eating fatty food.	We should avoid fatty food in our diet.

CLO #1: Utilize efficient writing style for producing an effective technical document.

Q2: Write a definitional paragraph on the topic "Deep Fakes". Imagine that you have collected the following information on this topic. Analyze the information, extract relevant points, synthesize and write the key points in your own words. Also, mention the sources of information. You do not need to add any information of your own and the word limit is 80 to 100 words. [10 marks]

[1] Deepfakes are the product of artificial intelligence (AI) applications that merge, combine, replace, and superimpose images and video clips to create fake videos that appear authentic. Deepfakes use recent advances in deep neural networks to create hyper-realistic synthetic media. When deepfake technology is used on videos or images, the face of a person can be swapped with another face leaving little trace of manipulation.

[2] Deepfakes are digitally manipulated synthetic media content (e.g., videos, images, sound clips) in which people are shown to do or say something- actions they never actually did in their lives. Advances in AI—particularly machine learning (ML) and deep neural networks (DNNs)—have contributed to the development of deepfakes.

[3] According to Korshunov and Marcel [5], deepfake is a tampering or manipulation technique that allows a user to swap the face of an individual, often an actor, actress or any other celebrity, with any other actor or person. Fake videos, audio or images are made that look and sound authentic. The system builds a model of a person saying or doing something by using large datasets containing recordings, videos or photos [6]. The manipulation is achieved via the use of hundreds or thousands of photos of the targeted person, which is used as a dataset [1].

----------The End----------